

10th International Conference of Information and Communication Technology (ICICT-2020)

Design and analysis of terminal live installation tool for robot in distribution work

Jinlu Zhang^a, Yifei Hu^a, Shanjun Xu^{b,*}, Tong Zhang^b, Shunan Ren^b, Mingrui Wang^c

^aState Grid Tianjin electric power company, Tianjin, China

^bBeijing Guodianfutong Science and Devel. Beijing, China

^cShandong Reapdaro Automation Technology Co., Ltd., Jinan, Shandong Province, China

Abstract

Live working in distribution work is an important way to ensure normal power supply of power equipment. Live installation tool for robot operation was proposed in this paper in order to reduce the safety accidents occurrence. Through different structures and sensor combinations, it can complete the autonomous operation by cooperating with robot, including lead wire monitoring, line detection and tightening the puncture clamp. Finally, the finite element analysis of gear transmission was carried out, and the results showed that the design of the tool can meet the strength requirements.

© 2021 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Peer-review under responsibility of the scientific committee of the 10th International Conference of Information and Communication Technology.

Keywords: Live working robot; Live wire connection; Tightening mechanism; Finite element analysis

1. Introduction

Live working in distribution work is a very important means in this respect of power supply and power grid safety [1,3]. At present, method of insulated blade and insulating gloves were widely used for manual operation. Insulated bucket truck was extensive used as the working platform in 10kV live working distribution work, where the operators completed the electrified work standing on the platform.

However, many problems were showed as followings [2,3,4]:

(1) High work intensity, low efficiency, and low automation level;

* Corresponding author. Tel.: +86-15153153836

E-mail address: 892856420@qq.com

- (2) Contact with high-voltage wire, high risk, personal electric shock and falling from high altitude;
- (3) Short circuit current and safety accident.

In order to solve the safety problems, a fully automatic live installation tool was developed in this paper, which could replace manual work and reduce the labor intensity. As a result, it could protect the safety of operators to the greatest extent.

2. Overall design

The live installation tool (hereinafter referred to as the tool) mainly realized the functions of grasping the puncture clamp, threading, and clamping the branch line, tightening the power output and other independent operation. The robot related to the tool through a flange and communicates by wireless way. The tool had the advantages of stable and reliable communication function and strong anti-electromagnetic interference ability.

2.1. Overall architecture

The tool mainly included clamp clamping mechanism, nut breaking mechanism, branch guiding mechanism, lifting mechanism and branch line position detection sensor. The overall structure was shown as Fig. 1 below.

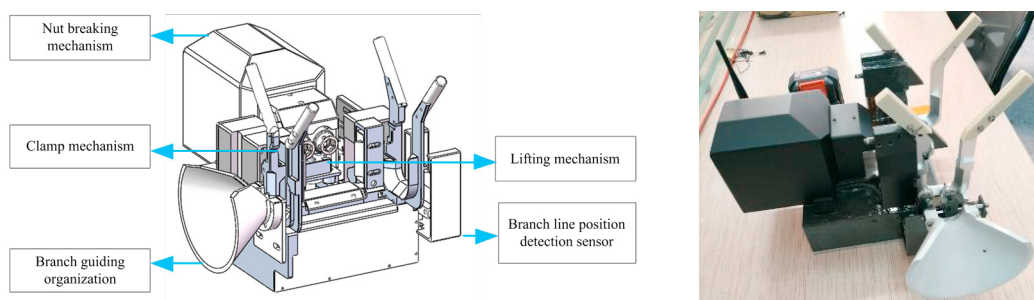


Fig. 1. Overall structure of the tool.

2.2. Nut tightening mechanism

The tightening mechanism was shown in Fig.2, which mainly includes the reducer, gear drive and motor drive.

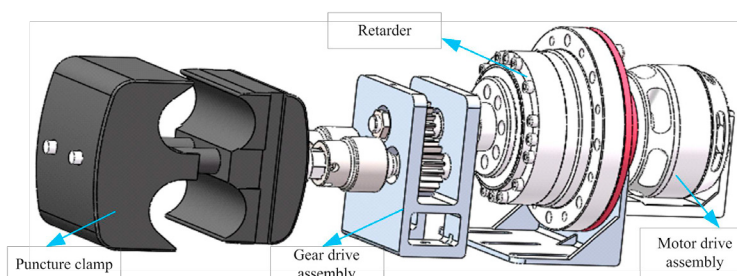


Fig. 2. Tightening mechanism.

Driven by the motor, the reducer made the driving gear shaft and driving gear rotate, in other words it drive the two gears to rotate. The two gears drive the tightening sleeve to rotate through the transmission shaft, and then screw off the torque nut of the puncture clamp to complete the electric connection.

The power is divided into two parts by the gear drive assembly. Gear and transmission shaft are made of 45 steel and then modulated.

The maximum torque was 28.6 N·m through the testing, which was as shown in Fig.3. It provided data support for gear transmission check.



Fig. 3. Torque test.

2.3. Drive gear transmission check

To carry out the strength check, a pair of transmission gears in gear transmission components were taking as an example, which provided reference for the design and verification of other parts on the tool.

2.3.1. Main technical parameters of gear transmission

Table 1. Main technical parameters of gear transmission components.

Item	Big gear wheel	Pinion
Type	Standard spur gear	Standard spur gear
Material	45 steel	45 steel
Teeth number z	41	17
Modulus m	1	1
Tooth width b	10mm	10mm
Pressure angle α	20°	20°

2.3.2. Calculation of coincidence degree

According to the mechanical design, the calculation formula was as follows [5,6,7]:

$$\begin{cases} d_b = mz \cos \alpha \\ d_a = mz + 2h_a m \end{cases} \quad (1)$$

The calculation formula of the pressure angle of the addendum circle was as follows [6,7]:

$$\begin{cases} \alpha_{a1} = \arccos \frac{d_{b1}}{d_{a1}} \\ \alpha_{a2} = \arccos \frac{d_{b2}}{d_{a2}} \end{cases} \quad (2)$$

The calculation formula of coincidence degree was as follows [6,7] :

$$\varepsilon_{\alpha} = \frac{z_1(\tan \alpha_{a1} - \tan \alpha) \pm z_2(\tan \alpha_{a2} - \tan \alpha)}{2\pi} \quad (3)$$

In formula (1) ~ (3), h_a was addendum height coefficient; d_b was diameter of base circle, mm; d_a was addendum circle diameter, mm; α_{a1}, α_{a2} were the pressure angle of the addendum circle of the two gear respectively, °; ε_{α} was coincidence degree, in the formula "+" corresponds to external meshing and "-" corresponds to internal meshing. This study took "+".

We could get the following results by introducing the parameters into the above formula:

$$\alpha_{a1} = 30.53^\circ, \alpha_{a2} = 25.36^\circ, \varepsilon_{\alpha} = 1.67$$

2.3.3. Traditional theoretical calculation of gear contact

The shape of the tooth surface of the gear was involute, and the width of the contact area was far less than the curvature radius of the contact point of the two tooth profiles. The maximum contact stress could be solved according to Hertz formula, which was shown as follows [5]:

$$\sigma_H = \sqrt{\frac{2KT(u+1)}{ubd_1^2}} Z_E Z_H \varepsilon_{\alpha} \quad (4)$$

Where, Z_E was the elastic influence coefficient, the calculation formula was as follows considering the influence of material elastic modulus and Poisson's ratio [5]:

$$Z_E = \sqrt{\frac{1}{\pi \left(\frac{1-\mu_1^2}{E_1} + \frac{1-\mu_2^2}{E_2} \right)}} \quad (5)$$

Where, E_1, E_2, μ_1, μ_2 were the elastic modulus (MPa) and poisson's ratio, in this paper, $E_1 = E_2 = 2.06 \times 10^{11}$ Pa, $\mu_1 = \mu_2 = 0.3$; Z_H was the area coefficient; for spur gear, Z_H was 2.5; ε_{α} was the coincidence coefficient; T was the transmitted torque, N·mm; d_1 was the diameter of the pinion indexing circle, mm; K was the load factor; u was the teeth ratio.

Since the coincidence coefficient was greater than 1 and less than 2, in order to facilitate the loading for the finite element model, and on the other hand, in order to compare the theoretical solution with the finite element solution, the load factor K and coincidence coefficient are both taken as 1. The maximum contact stress of spur gear pair is calculated as follows: $\sigma_H = 245$ MPa.

2.4. Finite element analysis

In this paper, the contact analysis of the gear of the robot's decelerating mechanism was carried out by using ANSYS. It was mainly divided into the following stages: establishing finite element model, meshing, applying load constraint, solving calculation and post-processing to check the results [8-10]. The process was as Fig.4, Fig.5 and Fig. 6.

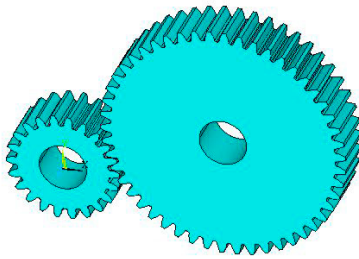


Fig. 4. Simplified 3D model.

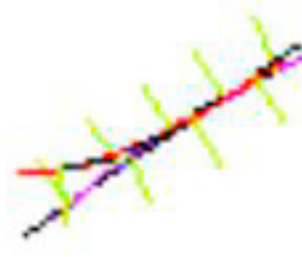


Fig. 5. Contact unit.

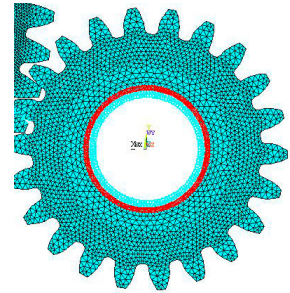


Fig. 6. Constraint and load of pinion.

The middle surface of three-dimensional cylindrical spur gear is extracted to establish a plane model and given a certain thickness^[10]. And plane 183 was selected. Simplified 3D model was shown in Fig. 4.

The quadrilateral is used for mesh generation^[10], and the size is 0.2mm. After division, 28046 nodes and 8941 quadrilateral elements were generated, which was shown in Fig. 5.

In the model of Fig.6, the inner surface of the big gear was set to be fully constrained^[9]. The axial degrees of freedom UX and radial degrees of freedom UZ of the inner surface of the pinion are constrained, and the rotational degrees of freedom (UY) of the pinion around the central axis of rotation were released.

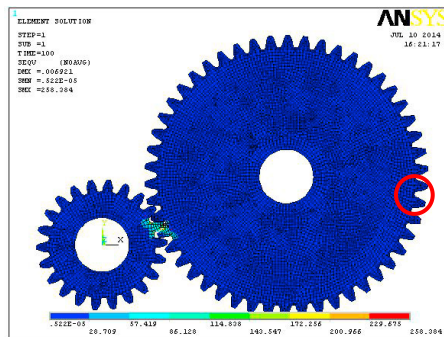
The circumferential force FY was applied on the inner surface of the pinion, and the calculation formula was^[6,7]:

$$F_Y = T / (R N) \quad (6)$$

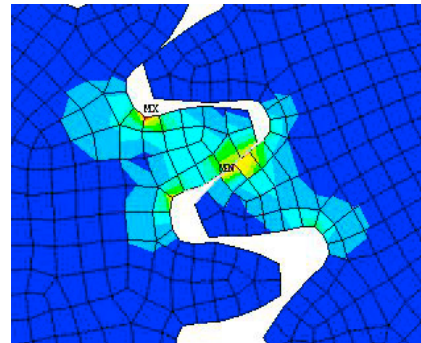
Where, T was the torque transmitted by the pinion, N·mm, $T = 9559P/n$, and n was the pinion speed; R was the radius of the inner ring of the pinion, mm; N was the number of inner ring nodes of gear.

According to the formula, $F_Y = 13\text{N}$ was calculated. Then the force was load to model as shown in Fig. 6.

Finally, the stress nephogram of the gear was obtained, as shown in Fig. 7.



a. Stress nephogram of gear



b. Enlarged view of contact area

Fig. 7. Stress nephogram of gear.

According to the calculation results in the above figure, the maximum contact stress was 258.384mpa, which was greater than the theoretical calculation value of 245MPa, and both were less than 550MPa of contact fatigue strength limit of 45 steel. The error was 5.3%, which showed that it was reasonable to simplify the finite element calculation, and the necessary simplification had little influence on the analysis results.

2.5. Conclusion

The tool designed in this paper could meet the expected functional requirements. The tool could replace manual operation, and avoid the risk accidents caused by manual connection of drainage lines, which had a positive role in the transformation of distribution line operation mode in China. The test results showed that the system could meet the operation requirements. At the same time, the overall structure design, and the tightening transmission gear components design were reasonable. As a result, the gear strength meets the requirements.

It had accumulated design and process experience for the research and development of special tools for connecting and leading wires of live working robots on high voltage lines.

Acknowledgement

This paper was fund by Science and Technology Projects of State Grid Corporation Limited (SGTJBHOYJJS1902138)

References

1. Wang Kaijun, Ying Hong, Lu Shouyin, et al. Development background and technical trend of high voltage live working robot [J]. *Zhejiang electric power*, 2005(06):16-18+30.
2. Rogerio Sales Gonc, alves. A mobile robot to be applied in high-voltage power lines [J], *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 2015, 37(1): 349 – 359.
3. Li Shuai, Zhang Liming, Guo Xin, Zhang Jianhai, Zhang Tong, Tang Xiao. Special tool for connecting and leading wire of 10kV distribution work live working robot [J]. *Equipment management and maintenance*, 2020(04):106-107.
4. Wu Ke, Zhang Jianwei, Hua Dong, et al. Current situation, and development of high voltage live working robot [J]. *Electronic technology and software engineering*, 2018(9):97-98.
5. Li yunwang, Ge Shirong, Zhu Hua. Study on gear differential mechanism of rocker mobile robot [J]. *Robot*, 2009, 31(3):235-241.
6. Pu Lianggui. *Mechanical design (8th Edition)*. Beijing: Higher Education Press. 2006
7. Qiu huaixuan. *Mechanical design (4th Edition)*. Beijing: Higher Education Press. 2007.
8. Johnson KL. *Contact mechanics [M]*. Translated by Xu bingye, et al. Beijing: Higher Education Press, 1992.
9. Li Yongxiang, Bi Xiaoqin, Zhang junshun, etc. load bearing contact analysis of spur gear based on ANSYS [J]. *Mechanical science and technology*, 2009, 28(7):931-940.
10. Yang Shenghua. Finite element analysis of gear contact [J]. *Acta computational mechanics*, 2003, 20(2):138-143.